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Document No.	QP-018
Title	FOD Prevention
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Prepared By	Owner
Approved By	Jacob Honick, Owner
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QP-018 FOD Prevention

1. Purpose

Define the housekeeping, tool control, and part protection practices that prevent Foreign Object Debris/Damage (FOD) from compromising product quality or causing damage to equipment.

2. Scope

Applies to all manufacturing areas, machining operations, assembly, and packaging activities within the shop.

3. Responsibilities

Role	Responsibility
Shop Owner	Establish and enforce FOD prevention practices, conduct daily housekeeping inspections, maintain tool accountability, ensure parts are protected between operations and before packaging

4. References

- QF-026 — FOD Prevention Checklist
- QP-020 — Handling, Storage, Packaging and Shipping
- QP-007 — Final Inspection and Product Release
- AS9100 Rev D Clause 8.5.4 (Preservation)

- NAS 412 (FOD Prevention)

5. Definitions

- **FOD (Foreign Object Debris):** Any substance, article, or agent alien to a product or system that could potentially cause damage or contamination.
- **FOD (Foreign Object Damage):** Damage caused by FOD to product or equipment.
- **Tool Accountability:** The practice of accounting for all tools, hardware, and consumables used during operations to prevent items from being left in/on product.

6. Procedure

Daily Housekeeping

1. Begin each shift by verifying work areas are clean and free of debris, chips, rags, and loose hardware.
2. Complete QF-026 Daily FOD/Housekeeping Checklist at the start of each work day.
3. Clean machines (chip trays, T-slots, work-holding surfaces) before loading a new part or new job setup.
4. Remove chips and coolant from finished parts immediately after each machining operation.

Tool Control and Accountability

1. Before beginning any operation, inventory tools and hardware to be used at the workstation.
2. At the completion of each operation, account for all tools and hardware. Investigate any missing item before moving the part to the next operation.
3. Do not leave loose hardware (screws, bolts, clamps, shims) on machine tables or inside machine enclosures after job completion.

Hardware Accountability During Machining

1. When using multiple fixture bolts, clamps, or parallels, record the count at setup start and verify the count at teardown.
2. Report any missing hardware to the Shop Owner before releasing the part from the operation.

Part Protection Between Operations

1. Place in-process parts on designated clean racks or in protective containers when not being actively machined.
2. Cover or wrap precision-machined surfaces to prevent handling damage, contamination, or corrosion between operations.

3. Do not stack parts on top of each other without protective separation.

Inspection Before Packaging

1. Before packaging for shipment, visually inspect all part surfaces for chips, debris, burrs, or foreign material.
2. Use compressed air or clean wipes to remove any remaining contamination.
3. Document FOD inspection completion on the Job Traveler or packing documentation.

FOD Awareness

1. The owner is the sole shop operator and maintains awareness of FOD prevention and its contribution to product safety. AS9100 clause 8.5.4 is a guidance reference only.
2. Review FOD awareness annually. Document the review on QF-016 Training Record with topic "FOD Prevention Awareness."
3. The owner reviews FOD precautions when the work or applicable customer requirements change.

7. Records Generated

Form No.	Title	When Completed
QF-026	Daily FOD/Housekeeping Checklist	Daily at start of shift

8. Revision History

Rev	Date	Description	Approved By
A	2026-08-07	Initial release	Shop Owner

Revision B Approval

Approved 2026-11-05 by Jacob Honick, Owner. Standardize control metadata and annual review, remove fixed page counts and align applicable references and wording.